

Hermenegildo Jerónimo dos Santos

Full Stack Software Engineer

+351 937 412 155 • hermeny7@hotmail.com • Covilhã, Portugal • linkedin.com/in/hermenegildosantos • github.com/HermenySantos

SUMMARY

Full Stack Engineer with 5+ years shipping production systems across healthcare, enterprise, and immersive-experience domains. Generalist across backend, frontend, mobile, cloud, and AI integration — ramps fast, delivers under pressure. Current highlight: architected a backend serving 2,100 concurrent users across 12 global locations in 5 weeks, first time on the stack. Stack: TypeScript · React · Go · Python · Azure OpenAI · Node.js.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Go, Python, Dart

Frontend / Mobile: React, Next.js, React Native (Expo/Reanimated), Flutter, Three.js, Tailwind CSS, Material UI, Redux

Backend: Node.js, NestJS, FastAPI, Express, GraphQL, WebSockets, MQTT, WebRTC, RESTful APIs

AI / LLM: Azure OpenAI GPT-4o, OpenAI Realtime API, TTS/STT, LLM prompt engineering, human-in-the-loop systems

Data: PostgreSQL, MongoDB, Cosmos DB, Firebase, Redis, MySQL — TypeORM, Prisma, SQLAlchemy

Cloud & DevOps: Azure (Functions, App Service, OpenAI), AWS (EC2, S3, Lambda), Cloudflare Workers, Docker, GitHub Actions, CI/CD

EXPERIENCE

Full Stack Engineer | Dorier

Jun 2025 – Present

Geneva, Switzerland · Remote

Scopely Global Event — WebAR Backend

- Architected Azure Functions + Cosmos DB backend serving 2,100 concurrent users across 12 global locations in 5 weeks — first time on that stack
- Rate limiting, anti-abuse, GDPR-aligned data handling; React admin dashboard; shipped on time with zero incidents

PMI AI Moderator — Live Event AI (pmi-ai-moderator.azurewebsites.net)

- Built human-in-the-loop AI moderator for live PMI panels: FastAPI + React (Vite/TypeScript/Tailwind) + Three.js visualiser; Azure OpenAI GPT-4o generation + gpt-4o-mini-tts voice synthesis
- WebRTC relay for OpenAI Realtime API, WebSockets for audience broadcast, Docker multi-stage build, Azure App Service deployment; supports OpenAI and Azure OpenAI via abstract provider interfaces

UN Geneva Visitor Center — Institutional Experience (FPDN)

- Core contributor (~27% of repo activity): TypeScript monorepo (Audio Guide app, Docent Controller, CMS, BoH cloud services), Go microservices, MQTT real-time messaging, Quuppa indoor positioning
- Built proximity-flash React Native Reanimated component (40 tests) used simultaneously across all visitor devices during the 'Gathering' exhibition moment

Enterprise Automation — Galderma / Lightcraft

- Python CLI/GUI tool with Dropbox integration, Gmail notifications, OAuth2 auth, cross-platform GitHub Actions CI/CD producing standalone executables — in production

Project Development Manager / Full Stack Developer | NomadEngenuity

Apr 2024 – Apr 2025

Covilhã, Portugal · Hybrid

- Architected microservice platform (Next.js + NestJS + TypeScript) for Seezy eye care app — improved data processing by 35%, reduced API response times by 40% via query optimisation and caching
- Implemented GraphQL + RESTful APIs handling 1,000+ daily transactions; configured Azure CI/CD with zero-downtime deployments
- Built PharmaSee healthcare staffing front-end (React, WebSockets, advanced matching algorithms); 65% mobile usage, 85% test coverage

Software Developer | DC Tech

Mar 2023 – Jun 2023

Covilhã, Portugal · Part-time

- Built award-winning full-stack environmental disaster prevention prototype (React, Node.js, ML models for flood/wildfire prediction); secured incubator funding in Porto

Programming Teacher | Conservatório de Música da Covilhã

Oct 2022 – Jun 2023

Covilhã, Portugal · Part-time

- Designed curriculum (Scratch, HTML/CSS, JavaScript) for 100+ students; built browser-based code editor with real-time preview and gamification (85% engagement rate)

Software Developer | 2GF Innovation Technology

Jul 2021 – Oct 2022

Remote

- Built React/TypeScript interfaces for cross-border family platform (10,000+ MAU), WebRTC video communication, reusable component library (~40% new-feature time), Flutter/Dart offline-first mobile app, Firebase auth/realtime DB
- Developed Node.js middleware integrating SAP enterprise systems; ReactJS dashboards for operational metrics; MongoDB analytics schemas

SELECTED PERSONAL PROJECTS

[InvoFlow](#) — FastAPI · Next.js 14 · PostgreSQL · Azure Document Intelligence · Cloudflare R2 · Clerk · Docker

- Full-stack invoice/receipt management SaaS for Portuguese SMBs — OCR-powered data extraction, real-time IVA tracking, accountant-ready export; Docker Compose deployment with mock mode for local dev

[NexTool API](#) — TypeScript · Cloudflare Workers · Hono · Vitest (30 tests) · RapidAPI

- 13+ developer utility endpoints (QR codes, meta scraping, text analytics, hashing, IP geolocation, colour conversion, JSON tools) deployed globally on Cloudflare edge; monetized on RapidAPI marketplace

[Totem Proximity Flash](#) — React Native · Expo · TypeScript · Reanimated · MQTT · Quuppa

- Open-source extraction of the UN Geneva proximity feature: distance-to-animation pure-function mapping, 40 automated tests, full MQTT/Quuppa production code preserved in original/ folder

EDUCATION

MBA (ESG Focus) · University of Beira Interior Executive, Portugal

2024 – Present

BSc Computer Science · University of Beira Interior, Portugal · Final project: React + Flutter + Firebase RBAC badge system

2018 – 2021

LANGUAGES

Portuguese — Native | **English** — Full Professional Proficiency (spoken & written)